Math 261-001
Quiz 6 Solutions, October 9

1. **Final answer:** $\boxed{\pi(2-\sqrt{2})\ln 2}$.

   Use spherical coordinates

   $$x = \rho\sin\varphi\cos\theta, \qquad y = \rho\sin\varphi\sin\theta, \qquad z = \rho\cos\varphi,$$

   where φ is measured from the positive z-axis. The sphere inequalities give $1 \le \rho \le 2$. Since the cone is $z = \sqrt{x^2+y^2}$, we can see immediately that this gives $0 \le \varphi \le \pi/4$. Otherwise, plugging in the coordinates into x, y, z we get $\rho\cos\varphi = \rho\sin\varphi$. There is no azimuthal restriction. Therefore:

   $$\begin{cases} 1 \le x^2+y^2+z^2 \le 4, \\ z \ge \sqrt{x^2+y^2} \end{cases} \qquad \Longrightarrow \qquad \begin{cases} 0 \le \theta \le 2\pi, \\ 0 \le \varphi \le \pi/4, \\ 1 \le \rho \le 2. \end{cases}$$

   The integrand is ρ^{-3}, and the volument differential is $dV = \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$. Consequently,

   $$\begin{aligned} I &= \int_0^{2\pi}\int_0^{\pi/4}\int_1^2 \frac{\sin\varphi}{\rho}\,d\rho\,d\varphi\,d\theta \\ &= 2\pi\,[-\cos\varphi]_0^{\pi/4}\,[\ln\rho]_1^2 = 2\pi\left(1-\frac{1}{\sqrt{2}}\right)\ln 2 = \pi(2-\sqrt{2})\ln 2. \end{aligned}$$

   The result is positive, as expected; the inner sphere excludes the singularity at the origin.

2. **Final answer:** $\boxed{S = 4\pi}$.

   For the fly's path,

   $$\begin{aligned} \vec{r}(t) &= (\cos t, \sin t, 2+\cos t+\sin t), \\ \vec{r}\,'(t) &= (-\sin t, \cos t, \cos t-\sin t), \\ \|\vec{r}\,'(t)\| &= \sqrt{\sin^2 t+\cos^2 t+(\cos t-\sin t)^2} = \sqrt{1+(\cos t-\sin t)^2}. \end{aligned}$$

   The scent intensity along the path is

   $$\sigma(\vec{r}(t)) = \sqrt{1+(\cos t-\sin t)^2} = \|\vec{r}\,'(t)\|.$$

   For a scalar line integral, multiply the scalar field by the speed: $ds = \|\vec{r}\,'(t)\|\,dt$, not just dt. Thus

   $$\begin{aligned} S &= \int_0^{2\pi} \sigma(\vec{r}(t))\|\vec{r}\,'(t)\|\,dt \\ &= \int_0^{2\pi} \big(1+(\cos t-\sin t)^2\big)\,dt = \int_0^{2\pi}(2-\sin 2t)\,dt \\ &= \big[2t+\tfrac{1}{2}\cos 2t\big]_0^{2\pi} = 4\pi. \end{aligned}$$

   This is scent intensity weighted by distance traveled, not a time integral of scent intensity. Reversing the flight direction would not change this scalar line integral.

3. **Final answer:** $\boxed{-2}$.

The path follows increasing t from $(\pi/2, \pi^2/4)$ to (π, π^2). Substitute $x = t$, $y = t^2$ into both components of the field:

$$\vec{F}(\vec{r}(t)) = \left(\frac{t^2 \cos t}{t^2}, -\frac{\sin t}{2t}\right) = \left(\cos t, -\frac{\sin t}{2t}\right), \qquad \vec{r}\,'(t) = (1, 2t).$$

Here $t \geq \pi/2 > 0$, so the denominators never vanish. The vector line integral uses the dot product with $\vec{r}\,'(t)$:

$$\begin{aligned}\int_C \vec{F} \cdot d\vec{r} &= \int_{\pi/2}^{\pi} \vec{F}(\vec{r}(t)) \cdot \vec{r}\,'(t)\, dt \\ &= \int_{\pi/2}^{\pi} (\cos t - \sin t)\, dt = [\sin t + \cos t]_{\pi/2}^{\pi} = -1 - 1 = -2.\end{aligned}$$

Do not insert an additional speed factor: $d\vec{r} = \vec{r}\,'(t)\, dt$. Reversing the orientation would give $+2$, which is not the requested integral.

4. **Final answers:**

Square	Field relative to the oriented tangent	Contribution
A	Opposite direction	Negative ($-$)
B	Perpendicular	Zero (0)
C	Same direction	Positive ($+$)

For the oriented unit tangent $\vec{T}$, the infinitesimal contribution is $\vec{F} \cdot \vec{T}\, ds$. The path's filled arrowheads determine $\vec{T}$; the thinner dashed arrows represent $\vec{F}$. Opposite, perpendicular, and matching directions give negative, zero, and positive dot products, respectively. These alignments hold throughout each highlighted piece, so the signs also describe the integrals over those pieces.

